

Atomic Force Microscopy

HOW DOES THE AFM WORK?

AFM provides a 3D profile of the surface on a nanoscale, by measuring *forces* between a sharp probe (<10 nm) and surface at very short distance (0.2-10 nm probe-sample separation). The probe is supported on a flexible cantilever. The AFM tip “gently” touches the surface and records the small force between the probe and the surface.

How are Forces Measured?

The probe is placed on the end of a cantilever (which one can think of as a spring). The amount of force between the probe and sample is dependant on the *spring constant* (stiffness) of the cantilever and the distance between the probe and the sample surface. This force can be described using Hooke's Law:

$$F = -k \cdot x$$

F = Force

k = spring constant

x = cantilever deflection

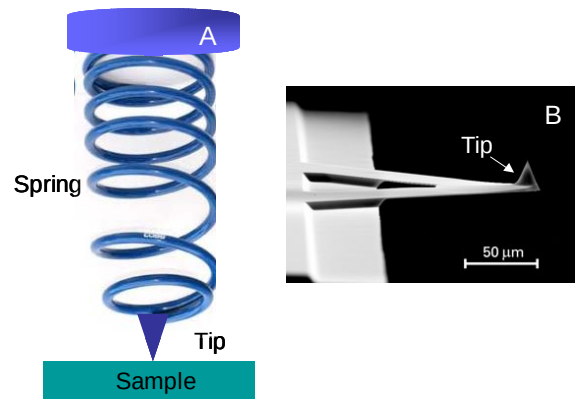


Figure 1. a) Spring depiction of cantilever b) SEM image of triangular SPM cantilever with probe (tip). (Image from [MikroMasch](#))¹

If the spring constant of cantilever (typically ~ 0.1-1 N/m) is less than surface, the cantilever bends and the deflection is monitored.

This typically results in forces ranging from nN (10^{-9}) to μ N (10^{-6}) in the open air.

What are probes made of?

Probes are typically made from Si_3N_4 , or Si. Different cantilever lengths, materials, and shapes allow for varied spring constants and resonant frequencies. A description of the variety of different probes can be found at various vendor sites.² Probes may be coated with other materials for additional SPM applications such as chemical force microscopy (CFM) and magnetic force microscopy (MFM).

Instrumentation

The motion of the probe across the surface is controlled similarly to the STM using *feedback loop* and *piezoelectric scanners*. (See [STM basic theory](#)) The primary difference in instrumentation design is how the forces between the probe and sample surface are monitored. The deflection of the probe is typically measure by a *“beam bounce” method*. A semiconductor diode laser is bounced off the back of the cantilever onto a position sensitive photodiode detector. This detector measures the bending of cantilever during the tip is scanned over the sample. The measured cantilever deflections are used to generate a map of the surface topography.

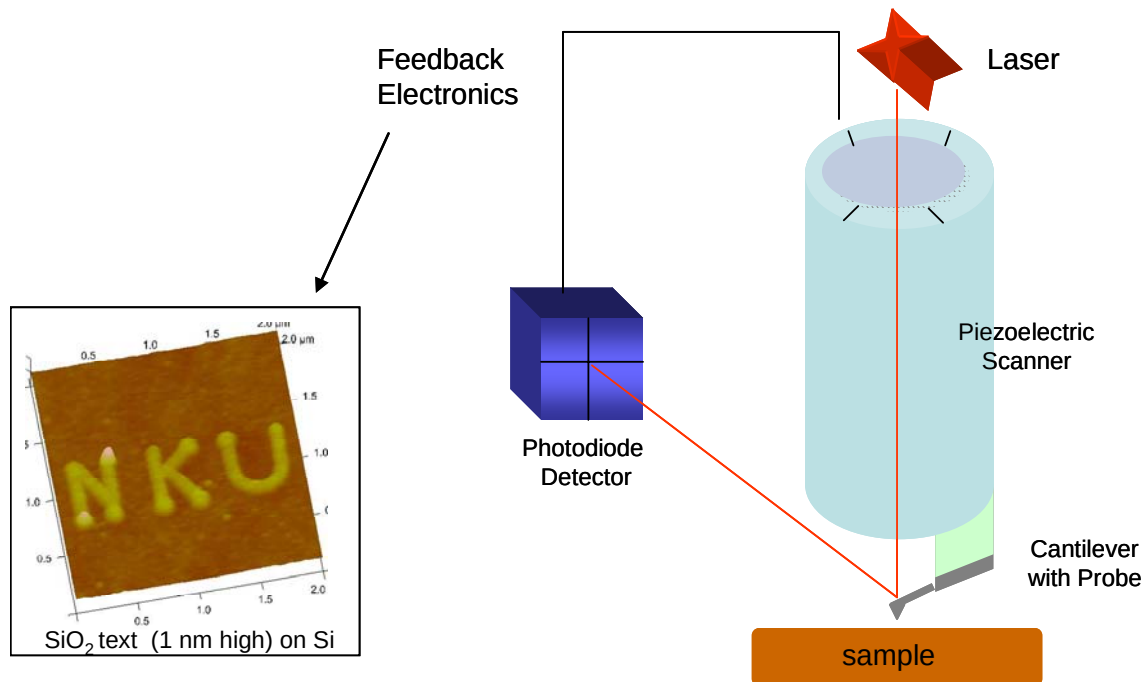


Figure 2. Schematic of AFM instrument showing “beam bounce” method of detection using a laser and position sensitive photodiode detector.

For a visual depiction of the “beam bounce” method of detection in AFM and you can refer to the following [web resource](#) which utilizes Legos ®, magnetics, and a laser pointer to demonstrate this concept.

Imaging Methods

What types of forces are measured?

The dominant interactions at short probe-sample distances in the AFM are Van der Waals (VdW) interactions. However long-range interactions (i.e. capillary, electrostatic, magnetic) are significant further away from the surface. These are important in other SPM methods of analysis.

During contact with the sample, the probe predominately experiences *repulsive Van der Waals forces* (contact mode). This leads to the tip deflection described previously. As the tip moves further away from the surface *attractive Van der Waals forces* are dominant (non-contact mode).

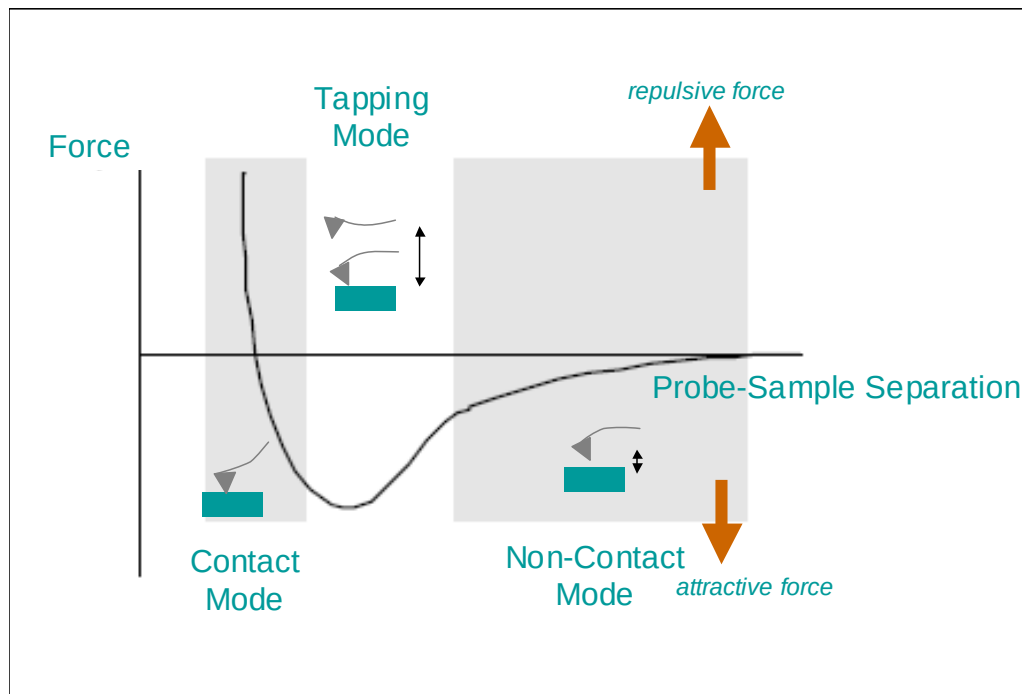


Figure 3. Plot of force as a function of probe-sample separation.

Modes of Operation

There are 3 primary imaging modes in AFM:

- (1) *Contact AFM*
< 0.5 nm probe-surface separation
- (2) *Intermittent contact (tapping mode AFM)*
0.5-2 nm probe-surface separation
- (3) *Non-contact AFM*
0.1-10 nm probe-surface separation

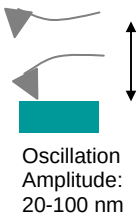
Primary Modes of Imaging:



1. Contact Mode AFM: (*repulsive VdW*) When the spring constant of cantilever is less than surface, the cantilever bends. The force on the tip is repulsive. By maintaining a constant cantilever deflection (using the feedback loops) the force between the probe and the sample remains constant and an image of the surface is obtained.

Advantages: fast scanning, good for rough samples, used in friction analysis

Disadvantages: at time forces can damage/deform soft samples (however imaging in liquids often resolves this issue)

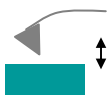
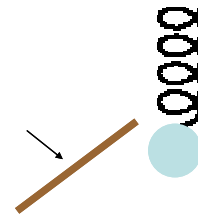


2. Intermittent Mode (Tapping): The imaging is similar to contact. However, in this mode the cantilever is oscillated at its *resonant frequency*, Figure 4. The probe lightly “taps” on the sample surface during scanning, contacting the surface at the bottom of its swing. By maintaining a constant oscillation amplitude a constant tip-sample interaction is maintained and an image of the surface is obtained.

Advantages: allows high resolution of samples that are easily damaged and/or loosely held to a surface; Good for biological samples

Disadvantages: more challenging to image in liquids, slower scan speeds needed

Figure 4. Resonant frequency is a natural frequency of vibration determined by the physical parameters of the vibrating object. For example if you hit a spring with a mass at the end (probe) the main response will be a bob up and down at its natural frequency.



3. Non-contact Mode: (*attractive VdW*) The probe does not contact the sample surface, but oscillates above the adsorbed fluid layer on the surface during scanning. (*Note: all samples unless in a controlled UHV or environmental chamber have some liquid adsorbed on the surface*). Using a feedback loop to monitor changes in the amplitude due to attractive VdW forces the surface topography can be measured.

Advantages: VERY low force exerted on the sample(10^{-12} N), extended probe lifetime

Disadvantages: generally lower resolution; contaminant layer on surface can interfere with oscillation; usually need ultra high vacuum (UHV) to have best imaging

What are Force Curves?

Force curves measure the amount of force felt by the cantilever as the probe tip is brought close to - and even indented into - a sample surface and then pulled away. In a force curve analysis the probe is repeatedly brought towards the surface and then retracted, Figure 5. Force curve analyses can be used to determine chemical and mechanical properties such as adhesion, elasticity, hardness and rupture bond lengths.

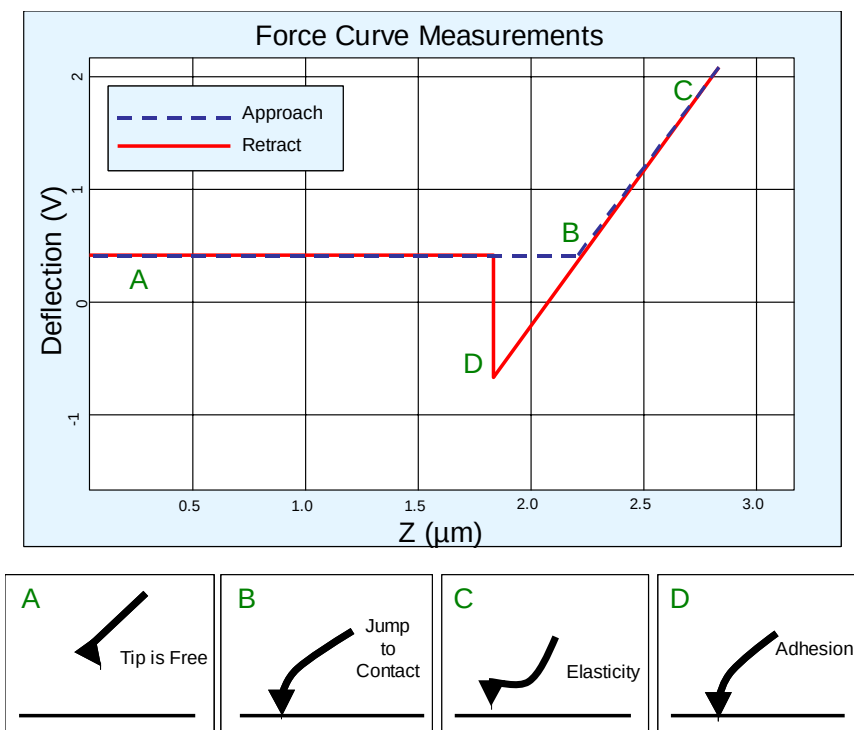


Figure 5. A typical force curve showing the various probe-sample interactions.

The slope of the deflection (C) provides information on the hardness of a sample. The adhesion (D) provides information on the interaction between the probe and sample surface as the probe is trying to break free. Direct measurements of the interactions between molecules and molecular assemblies can be achieved by functionalizing probes with molecules of interest ([see Chemical Force Microscopy](http://www.ntmdt.ru/SPM-Techniques/Principles/Spectroscopies/Force-distance_curves_mode20.html)).

Interactive Tools:

An interactive force curve analysis can be found here: http://www.ntmdt.ru/SPM-Techniques/Principles/Spectroscopies/Force-distance_curves_mode20.html

WHAT ARE THE LIMITATIONS OF AFM?

The AFM can be used to study a wide variety of samples (i.e. plastic, metals, glasses, semiconductors, and biological samples such as the walls of cells and bacteria). *Unlike STM or scanning electron microscopy it does not require a conductive sample.* However there are limitations in **achieving atomic resolution**. The physical probe used in AFM imaging is not ideally sharp. As a consequence, an AFM image does not reflect the true sample topography, but rather represents the interaction of the probe with the sample surface. This is called **tip convolution**, Figure 6.

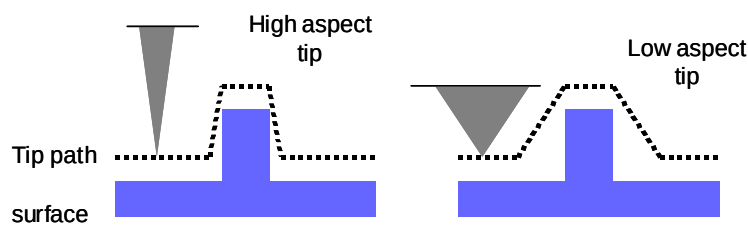


Figure 6. Ideally a probe (tip) with a high aspect ratio will give the best resolution. The radius of curvature of the probe leads to tip convolution. This does not often influence the height of a feature but the lateral resolution.

Commercially available probes are becoming more widely available that have very high aspect ratios. These are made with materials such as carbon nanotubes or [tungsten spikes](#).³ However these probes are still very expensive to use for every day image analysis.

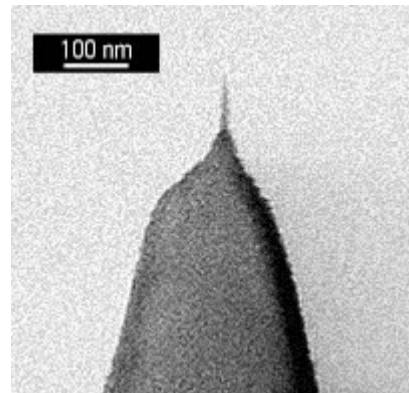


Figure 7. Example of High aspect probes from MikroMasch (1nm radius of curvature HI'RES-W probe).³

Online Resources

Another useful source on the principles of SPM:

http://www.physics.leidenuniv.nl/sections/cm/ip/group/Principle_of_SPM.htm

Another useful source on how AFM works

<http://stm2.nrl.navy.mil/how-afm/how-afm.html#General%20concept>

A review of atomic force microscopy imaging systems

Abstract

The atomic force microscope (AFM) system has evolved into a useful tool for direct measurements of micro-structural parameters and unraveling the intermolecular forces at nanoscale level with atomic-resolution characterization. Typically, these micro-cantilever systems are operated in three open-loop modes; non-contact mode, contact mode, and tapping mode. In order to probe electric, magnetic, and/or atomic forces of a selected sample, the non-contact mode is utilized by moving the cantilever slightly away from the sample surface and oscillating the cantilever at or near its natural resonance frequency. Alternatively, the contact mode acquires sample attributes by monitoring interaction forces while the cantilever tip remains in contact with the target sample. The tapping mode of operation combines qualities of both the contact and non-contact modes by gleaming sample data and oscillating the cantilever tip at or near its natural resonance frequency while allowing the cantilever tip to impact the target sample for a minimal amount of time. Recent research on AFM systems has focused on many manufacturing and metrology processes at molecular levels due to its tremendous surface microscopic capabilities. This paper provides a review of such recent developments in AFM imaging system with emphasis on operational modes, micro-cantilever dynamic modeling and control. Due to the important contributions of AFM systems to metrology and biological sciences, this study also provides a comprehensive review of recent applications of different AFM systems in these two important areas.

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1. Introduction

The atomic force microscope (AFM) system has evolved into a useful tool for direct measurements of intermolecular forces with atomic-resolution characterization that can be employed in a broad spectrum of applications such as electronics, semi-conductors, materials and manufacturing, polymers, biology and biomaterials. AFM provides additional capabilities and advantages relative to other microscopic methods (e.g. scanning electron microscopy (SEM) and transmission electron microscopy (TEM)) in studies of metallic surfaces and micro-structures by providing reliable measurements at the nanometer scale [37,57,75,121,123]. AFM can also be used for nanoindentation to provide *in situ* imaging ability without moving the sample, switching tips, relocating the area for scanning, or using an entirely different instrument to image the indentation [35,56,71,102,118]. Force modulation microscopy (FMM) which is an extension of AFM imaging, is used extensively for the characterization of mechanical properties and in applications such as imaging composition changes in a composite material, analyzing polymer homogeneity, and detecting contaminants in manufacturing processes [3,27,99]. Assembly of nano-particles and linking them to electrical leads, such as random deposition of clusters between electrodes, binding by wet chemistry, and electrostatic trapping, all serve as other important applications of the AFM technique.

As depicted in Fig. 1, a typical AFM system consists of a micro-machined cantilever probe and a sharp tip mounted to a Piezoelectric (PZT) actuator and a position sensitive photo detector for receiving a laser beam reflected off the end-point of the beam to provide cantilever deflection feedback. The principle of AFM operation is to scan the tip over the sample surface with feedback mechanisms that enable the

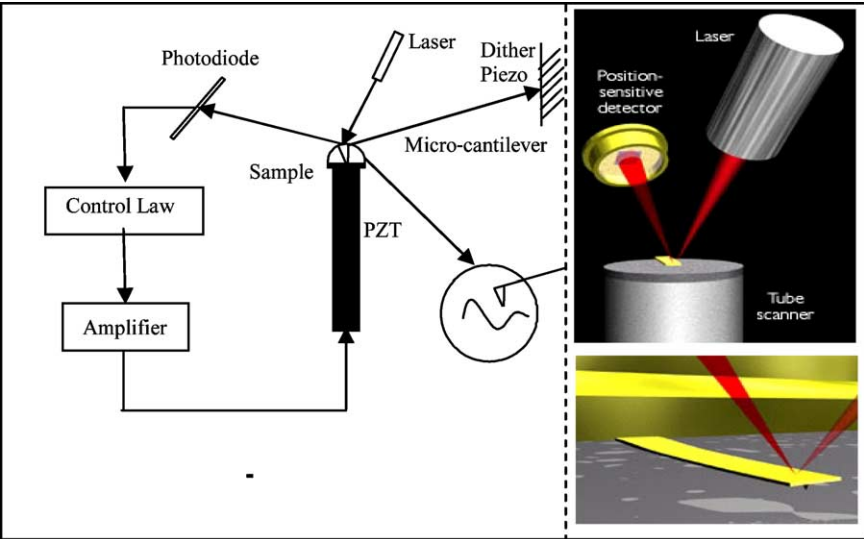


Fig. 1. Schematic of basic AFM operation (left), real micro-cantilever and components (right).

PZT scanners to maintain the tip at a constant force, or constant height above the sample surface. As the tip scans the surface of the sample, moving up and down with the contour of the surface, the laser beam deflected from the cantilever provides measurements of the difference in light intensities between the upper and lower photo detectors. Feedback from the photodiode difference signal, through software control from the computer, enables the tip to maintain either a constant force or constant height above the sample. In the constant force mode, the PZT transducer monitors real time height deviation. In the constant height mode, the deflection force on the sample is recorded.

The rest of the paper is organized as follows. In the immediately following section, the AFM operation and control modes are discussed. Three open-loop modes, namely non-contact, contact and tapping modes, are described. Section 3 provides a detailed discussion on the AFM modeling and presents both lumped-parameters and distributed-parameters modeling schemes. Some recent applications of AFM imaging systems in both molecular metrology and biological sciences are presented in Section 4, followed by the concluding remarks in Section 5.

2. AFM operational and control modes

Typically, these micro-cantilever systems are operated in three open-loop modes: (i) non-contact mode, (ii) contact mode, and (iii) tapping mode. In order to probe electric, magnetic, and/or atomic forces of a selected sample, the non-contact mode is utilized by moving the cantilever slightly away from the sample surface and oscillating the cantilever at or near its natural resonance frequency. By mounting the cantilever on a PZT element and measuring the shift from its natural resonance frequency due to sample attractive interactions, topographical information of the sample can be extracted [12]. Alternatively, the contact mode acquires sample attributes by monitoring interaction forces while the cantilever tip remains in contact with the target sample [34]. The tapping mode of operation combines qualities of both the contact and non-contact modes by gleaning sample data and oscillating the cantilever tip at or near its natural resonance frequency while allowing the cantilever tip to impact the target sample for a minimal amount of time [95,101].

Though widely practiced, open-loop operation modes of AFM exhibit the potential for chaotic behavior in the cantilever tip displacement thus rendering low resolution topographical information. As a result, recent research on AFM systems has focused on detailed numerical analysis such that this chaotic behavior region can be well defined and ideally avoided [12,101]. In addition to analytical methodology, several feedback control strategies have been developed in order to improve the AFM region operation [9,47].

The interaction forces between the tip and sample in all of the three open-loop modes can be distinctly identified on a force–displacement curve as shown in Fig. 2. When the interatomic distance is quite large, weak attractive forces are generated between the tip and the sample. As the atoms are gradually brought closer to each other, the attractive forces increase until the atoms become so close that the electron

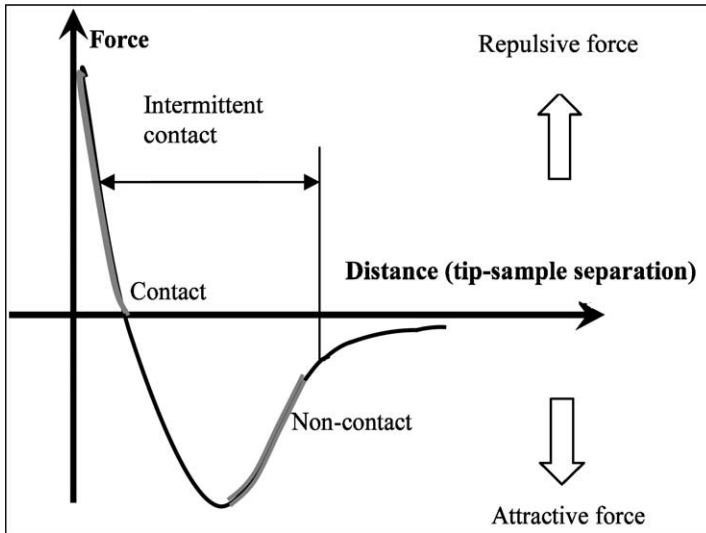


Fig. 2. Interatomic force variation versus distance between AFM tip and sample.

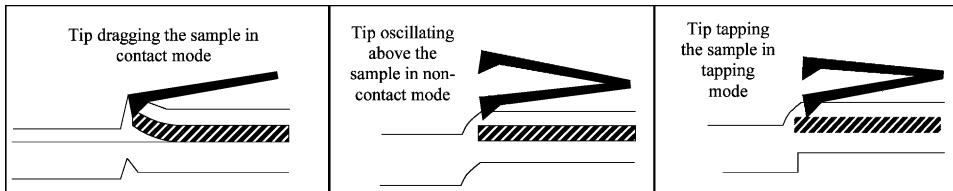


Fig. 3. Contact mode (left), non-contact mode (middle) and tapping mode (right).

clouds begin to repel each other electrostatically. This repulsive force between the atoms progressively weakens the attractive forces as the interatomic distance decreases. The interaction force becomes zero when the distance between the atoms reaches a couple of Angstroms and becomes fully repulsive when the atoms are in contact (see Fig. 2).

Fig. 3 depicts a schematic diagram of the nature of operation in all the three modes as well as the surface topography that would be generated in each mode. It can be seen that the tapping and non-contact modes reveal the surface topography of the sample to a greater extent when compared to the contact mode. Also, the operation of the AFM in contact mode damages the sample surface due to the lateral dragging forces exerted by the probe tip. The detailed description and operational characteristics of each mode is described next.

2.1. Non-contact AFM (NC-AFM) imaging systems

In this mode, the cantilever tip hovers about 50–150 Å above the sample surface to detect the attractive van der Waals forces acting between the tip and the sample,

and topographic images are constructed by scanning the tip above the surface. Since the attractive forces from the sample are substantially weaker than the forces used by contact mode, the tip must be given a small oscillation so that these small forces can be detected by measuring the change in amplitude, phase, or frequency of the oscillating cantilever. Current practice calls for detection of an amplitude shift that occurs when the tip is in the van der Waals regime. In many cases, the fluid contaminant layer is substantially thicker than the range of the van der Waals force gradient, and therefore, attempts to image the true surface with NC-AFM fail as the oscillating probe becomes trapped in the fluid layer or hovers beyond the effective range of the forces it attempts to measure. This is a major drawback of the NC-AFM imaging technique, which significantly degrades the resolution of the generated topographical images. This *barely contact mode* in practice is used to determine a set point (where tip touches sample), around which a feedback loop is utilized to control the sample positioning with respect to the tip. Another major shortfall of the NC-AFM systems is the lack of measurement of the distance between sample surface and micro-cantilever tip, i.e. z_0 in Fig. 4 (left). For highest resolution, it is necessary to measure this distance which will ultimately result in improved interaction force estimation and enhanced topographical images.

Since its invention [1,68], the NC-AFM has been increasingly used for imaging a wide class of materials, ranging from metals to semi-conductors and from polymers to biological materials [122]. The NC-AFM offers unique advantages over other contemporary scanning probe techniques such as contact AFM (C-AFM) and scanning tunneling microscopy (STM). Unlike STM and C-AFM, the absence of repulsive forces in NC-AFM permits the imaging of “soft” samples to provide topography with little or no contact between the tip and the sample. At the same time, the ability to resolve atomic rows, steps, or vacancies on flat (i.e. “two-dimensional”) semi-conductor surfaces using NC-AFM has brought this microscope almost on par with the STM [36,41,115]. NC-AFM is desirable because it provides a means for measuring sample topography with little or no contact between the tip and the sample. The total force between the tip and the sample in the non-contact regime is very low, generally about 10–12 pN. This low force is advantageous for studying soft or elastic samples. A further advantage is that samples like silicon wafers are not contaminated through contact with the tip. Therefore, it does not suffer from tip or sample degradation effects that are sometimes observed after taking numerous scans with C-AFM. Since NC-AFM imaging involves the tip hovering at a fixed distance away from the sample surface, the minimal tip–sample interaction yields improved resolution over C-AFM.

2.2. Contact AFM (C-AFM) imaging systems

In this mode, also known as *Repulsive mode*, the tip is in close contact with the sample as it scans the surface (see Fig. 5). In this case, the interaction forces between the tip and the sample are mainly repulsive in nature. The examination of the van der Waals curve, as seen in Fig. 2, plotted for varying interatomic distances reveals that the slope of the curve in the contact regime is very steep. This is because, at such

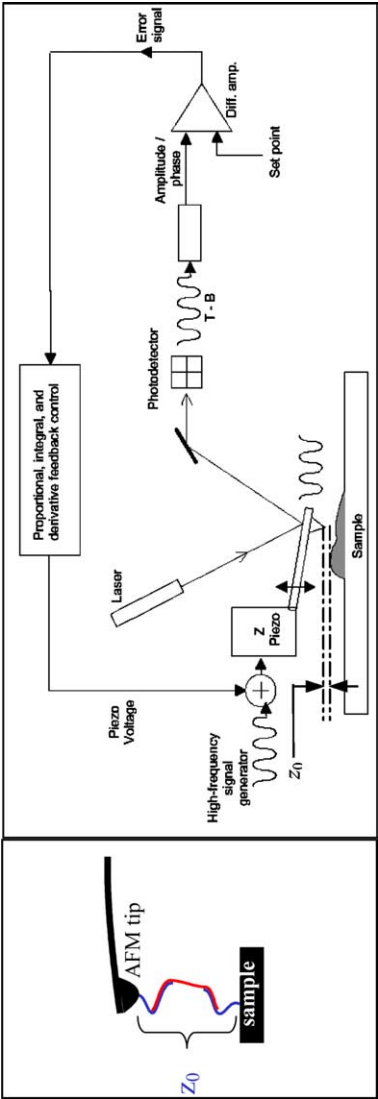


Fig. 4. Schematic diagram of non-contact AFM operation.

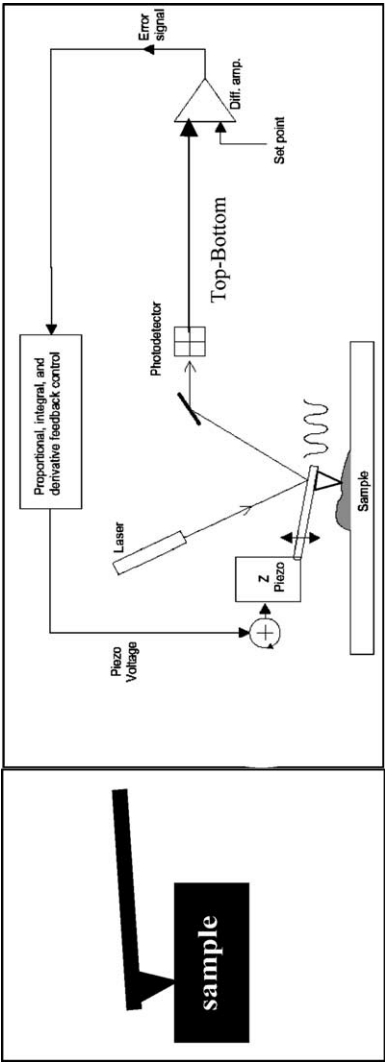


Fig. 5. Schematic diagram of contact AFM operation.

close interatomic distances, the electron clouds in the atoms repel each other electrostatically. As a result, the repulsive van der Waals force dominates any other attractive force that may tend to act.

In addition to these repulsive forces, there are two other forces generally present in C-AFM: capillary force due to the presence of a contaminant layer over the sample surface and the force exerted by the cantilever itself (this is the shear force component of the interaction force acting on the micro-cantilever). The capillary force is attractive in nature and remains constant throughout the sample (assuming that the contaminant layer is homogeneous). However, the magnitude and direction of the cantilever forces acting on the sample depend on the deflection and spring constant of the cantilever.

In C-AFM, the surface profile can be generated by operating in either of the two modes: *constant height* mode or *constant force* mode. In *constant height* mode, the PZT scanner holding the AFM tip laterally scans the surface of the sample without moving in the *z*-direction. The cantilever deflection thus produced as a result of the tip-sample interaction can be utilized to estimate the surface topography. The drawback of this technique is that the presence of any steep steps on the sample surface may damage the tip as it will be pushed against the surface irregularities during the scanning process.

In the *constant force* mode, which is most widely utilized, the normal force acting between the tip and the sample is kept constant through a feedback loop. A PZT positioning element is used to position the probe tip and apply the required force on the sample. Initially, the tip is brought in contact with the sample at a particular position until the required cantilever deflection is attained. Now, the tip is scanned laterally over the sample and the cantilever deflection is recorded by an optical detection system. The measured value of deflection is compared to a set value in a DC feedback amplifier and the resulting error signal from the amplifier is used to actuate the PZT positioning element by applying the required voltage. This, in turn, raises or lowers the cantilever to restore the desired deflection and thus maintains constant force (and thereby a constant height) between the tip and sample. The voltage applied by the feedback amplifier to the PZT element is a measure of the surface topography and is expressed as a function of the lateral position of the sample. This function is used to generate the surface topography of the specimen.

The main drawback of the contact mode is that the shear forces resulting from the lateral movement of the tip tend to damage soft samples and distort the features of the generated image. Therefore, this mode is not suited for examining of soft biological and polymer surfaces as it causes substantial sample degradation.

2.3. Tapping AFM (T-AFM) imaging systems

The technique of tapping mode AFM (T-AFM) is a key advance in AFM technology. This technique allows high resolution imaging of soft samples that are difficult to examine using the contact AFM technique. It overcomes problems such as friction and adhesion that are usually associated with conventional AFM imaging systems. In this mode, the cantilever is oscillated at or near its natural resonant

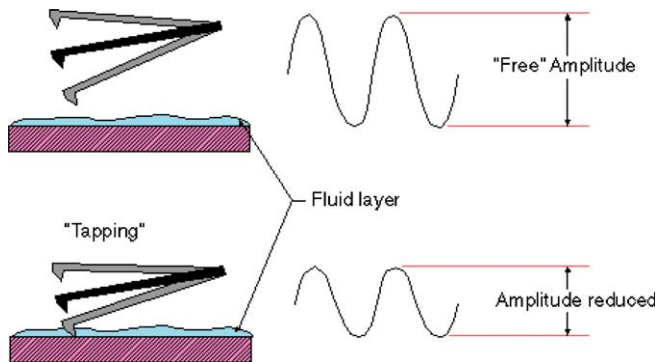


Fig. 6. Schematic of tapping mode AFM operation [87].

frequency using a PZT actuator (Fig. 6). The PZT actuator applies a force on the cantilever base and causes the cantilever tip to vibrate at amplitudes which are typically in the range of 20–100 nm when the tip is not in contact with the surface. The vibrating tip is now moved close to the sample until it begins to lightly tap the surface. During scanning, the probe tip alternately touches the surface and lifts off at frequencies of about 50–500 kcycles/s. Owing to the energy losses caused due to intermittent contacting of the tip with the surface, the amplitude of vibration changes according to the surface topography of the sample.

During tapping mode operation, the oscillation amplitude is kept constant through a feedback loop. When the tip passes over a bump in the surface, its vibration amplitude decreases due to availability of less vibrating space. On the other hand, when it passes over a depression, its vibration amplitude increases (approaching its free air amplitude). This change in oscillation amplitude is detected by the optical system and fed back to the controller which compares the measured value with the set reference value and generates an error signal. This signal actuates the PZT element that adjusts the tip–sample separation to maintain constant amplitude and thereby constant force on the sample. The error signal applied to the PZT, which is a measure of the surface irregularities in the vertical direction, is expressed as a function of the tip's lateral position and used to plot the surface topography of the sample.

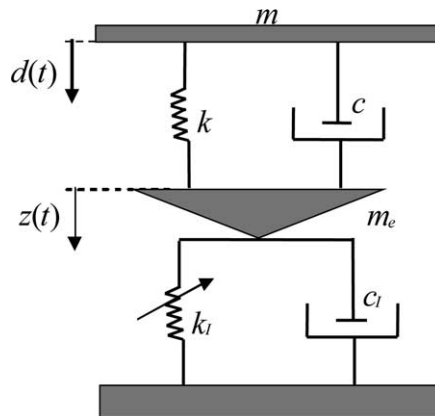


Fig. 7. Single-degree-of-freedom lumped model of a tapping mode AFM, [18].